Jacopo Grilli

Contact Information

Research Scientist
Abdus Salam International Centre for Theoretical Physics (ICTP)
Strada Costiera, 11
34151, Trieste, Italy

Work: +39-040-224-0148
grilli.jacopo@gmail.com
jacopogrilli.github.io

Vita

- November 2023 to present
  Research Scientist at Quantitative Life Sciences, ICTP, Trieste, Italy.
- May 2019 to November 2023
  Associate Research Officer at Quantitative Life Sciences, ICTP, Trieste, Italy.
- January 2018 to April 2019
  Omidyar Postdoctoral Fellow at Santa Fe Institute, Santa Fe, NM, USA.
- January 2015 to December 2017
  Postdoctoral Scholar at Department of Ecology and Evolution, University of Chicago, Chicago, IL, USA.
  Advisor: S. Allesina
- January 2012 to February 2015
  Ph.D. in Physics at Università degli Studi di Padova, Padova, Italy.
  Advisor: A. Maritan.
  Ph.D. awarded 27 February 2015
- October 2011 to December 2011
  Post-Master Scholarship 'ex 60%' 2011 at Department of Physics and Astronomy G. Galilei, Università degli Studi di Padova, Padova, Italy.
- October 2009 to July 2011
  M.S. in Theoretical Physics at Università degli Studi di Milano.
  Advisors: A. Maritan and B. Bassetti. Final grade *110/110 cum Laude.*
- October 2006 to October 2009
  B.S. in Physics at Università degli Studi di Milano.
  Advisors: B. Bassetti and M. Cosentino Lagomarsino. Final grade *110/110 cum Laude.*

Editor

Plos Computational Biology (Associate Editor, 2019-)

Oikos (Editorial board, 2018-)

Royal Society Open Science (Associate Editor, 2021-)

Plos Computational Biology (Guest Editor, 2018-2019)

Complexity (Special issue "Scales and Complexity in Ecological Communities: Models, Methods, and Predictions", 2018)

Reviewer

Grants: *European Research Council (EU)*, *National Science Foundation (USA)*, *Swiss National Science Foundation (Switzerland)*, *Israel Science Foundation (Israel)*

Journals: *Nature, Science, Nature Ecology and Evolution, Nature Communications, Science Advances, PNAS, eLife, Physical Review Letters, Plos Computational Biology, Physical Review X, Ecology Letters, The ISME Journal, American Naturalist, Proceedings of the Royal Society B, Proceedings of the Royal Society A, Journal of Statistical Mechanics, Journal of Statistical Physics, Physical Review E, Frontiers in Ecology and Evolution, Scientific Reports, Plos One, npj Systems Biology and Applications, Methods in Ecology and Evolution, Journal of Theoretical Biology, Oikos, Entropy, Journal of Biogeography, Journal of Complex Networks, Functional Ecology, Communications in Nonlinear Science and Numerical Simulation*
Publons ID 558637

Organized conferences and workshops

- May 18 - June 18, 2026
  KITP Program, Coarse-graining Microbial Ecology: from Genes to Physiological Strategies to Communities across Environments.
- September 8 - September 19, 2025
  ICTP Program, Hands-On School in Quantitative Biology.
- January 6 - January 17, 2025
  NCBS School, ICTP-ICTS Winter School on Quantitative Systems Biology.
- November 4 - November 15, 2024
  University of Havana School, Hands-On Quantitative Biology School.
- June 3 - June 7, 2024
  ICTP Conference, Annual meeting of the Physics of Living Systems (PoLS) Student Research Network.
- May 6 - May 10, 2024
  ICTP Workshop, TLQS Workshop on Limits to Collective Agency.
- February 19 - March 15, 2024
  ICTP School, Spring College on the Physics of Complex Systems.
- October 2 - October 13, 2023
  ICTP Workshop, Advanced School on Quantitative Principles in Microbial Physiology: from Single Cells to Cell Communities.
- April 17 - April 21, 2023
  ICTP Workshop, TLQS Workshop on Inequality Across Scales, Space, Time and Domains.
- February 20 - March 17, 2023
  ICTP School, Spring College on the Physics of Complex Systems.
- July 25 - July 29, 2022
  ICTP Workshop, Quantitative Human Ecology.
- June 6 - June 24, 2022
  ICTP Workshop, Eco-evolutionary Dynamics of Microbial Communities Across Scales.
- January 19 - January 21, 2021
  ICTP Workshop, Limits to Diversity Assembly.
- November 30 - December 18, 2020
  ICTP Winter School on Quantitative Systems Biology, Quantitative Approaches in Ecosystem Ecology.

- February 10-12, 2020
  Santa Fe Institute Working Group, Aging in Single-celled Organisms: from Bacteria to the Whole Tree of Life.
- January 20-25, 2020
  ICTP-SAIFR School, Community Ecology: from patterns to principles.
- March 4-6, 2019
  Santa Fe Institute Working Group, Higher-Order Interactions: Experiments, Inference and Models.
- January 29-31, 2019
  Santa Fe Institute Working Group, Irreversible Processes in Ecological Evolution.
- June 12, 2018
  NetSci 2018 Satellite Meeting, EcoNet. Ecological networks: spandrels, selection, and assembly.
- September 20, 2016
  CCS 2016 Satellite Meeting, Living 2.0. Robustness, Adaptability and Critical Transitions in Living Systems.
- September 16-19, 2015
  Workshop, Living Systems: from Interaction Patterns to Critical Behavior.
- September 25, 2014
  ECCS 2014 Satellite Meeting, LIVING. Robustness, Adaptability and Critical Transitions in Living Systems .

Seminars at Institutions

- February 12, 2025. Theory of Living Matter Seminar Series.
  Invited Seminar: *Microbial community dynamics in-silico, in-vitro, and in-vivo*
- April 18, 2024. Dutch Institute for Emergent Phenomena.
  Invited Seminar: *Competition without contingency and functional convergence*
- January 25, 2024. Slovenian Microbiological Society Online Seminars.
  Invited Seminar: *What is typical in microbial communities?*
- March 24, 2023. Population Dynamics Online Seminar Series.
  Invited Seminar: *What is typical in microbial communities?*
- February 7, 2023. NCCR Microbiomes Seminar Series, Switzerland.
  Invited Seminar: *What is typical in microbial communities?*
- November 24, 2021. Dept of Physics, Ecole Normale Supérieure, Paris, France.
  Invited Seminar: *What is typical in microbial communities?*
- October 5, 2021. Lecture Series in Ecology and Evolution, Institute of Ecology and Evolution, Universitat Bern, Switzerland.
  Invited Seminar: *What is typical in microbial communities?*
- October 5, 2021. Lecture Series in Ecology and Evolution, Institute of Ecology and Evolution, Universitat Bern, Switzerland.
  Invited Seminar: *What is typical in microbial communities?*
- April 14, 2021. Centre for Ecological Sciences, Indian Institute of Science, India.
  Invited Seminar: *What is typical in microbial communities?*
- March 19, 2021. Instituto Carlos I, university of Granada, Spain.
  Invited Seminar: *What is typical in microbial communities?*
- February 25, 2021. Biological Complexity Unit, Okinawa Institute of Science and Technology, Japan.
  Invited Seminar: *What is typical in microbial communities?*

- September 30, 2020. Department of Biology, Hong Kong Baptist University, Hong Kong.
  Invited Seminar: *Laws of diversity and variation in microbial communities*
- August 24, 2020. Dept. of Physics, University of Florida.
  Invited Seminar: *Laws of diversity and variation in microbial communities*
- April 21, 2020. Rockefeller university.
  Invited Seminar: *Laws of diversity and variation in microbial communities*
- February 4, 2019. CNLS Colloquium, Los Alamos National Lab.
  Invited seminar: *Higher-order interactions stabilize the dynamics of ecological communities*
- April 15, 2016. Laboratory of Computational and Quantitative Biology, UPMC.
  Invited Seminar: *Coexistence in large ecosystems: from structure to function*
- April 12, 2016. ICTP.
  Invited Seminar: *Coexistence in large ecosystems: from structure to function*
- May 26, 2015. Department of Ecology and Evolution, University of Chicago.
  Invited Seminar: *Stability and feasibility of large ecosystems*
- March 26, 2015. Wageningen University.
  Invited Seminar: *On the stability of large ecosystems*
- November 3, 2014. Department of Environmental Systems Science, ETH.
  Invited Seminar: *Spatial aggregation and spatial fragmentation: simple random models for spatial ecology*
- October 6, 2014. Dipartimento di Fisica, Università di Torino.
  Invited Seminar: *Scaling laws in genome evolution*
- December 17, 2013. University of Illinois at Urbana-Champaign.
  Invited Seminar: *Emergence of criticality in living systems through adaptation and evolution*

Talks at meetings

- July 22, 2025. Unifying theories in high-dimensional Biophysics, ICTS.
  Invited Talk: *Microbial ecology and evolution in high dimension*
- July 17, 2025. StatPhys29.
  Invited Talk: *High-dimensional microbial community dynamics*
- June 12, 2025. 20th CFGBC Symposium.
  Invited Talk: *Microbial community dynamics in-silico, in-vitro, and in-vivo*
- March 18, 2025. APS March Meeting 2025.
  Contributed Talk: *The statistical structure of alternative stable states in microbial communities*
- January 6 - January 17, 2025. ICTP-ICTS Winter School on Quantitative Systems Biology, NCBS.
  Lecturer: *Stochasticity and Randomness in Ecology*
- September 9, 2024. IICII Summer School in Quantitative Biology.
  Guest Lecture: *Who is the fittest? Non-trivial insights from simple models*
- August 30, 2024. Annual Meeting of the Swiss Society for Microbiology.
  Invited Talk: *Simplifying complex microbial community dynamics*
- July 1, 2024. IX Mediterrenean School of Complex Networks.
  Invited Talk: *Some unanswered and unasked questions in microbial ecology*

- June 10, 2024. Workshop on Naturam totam complectari animo: Towards a relational ecology, Pontificial University of the Holy Cross.
  Invited Talk: *What is a macroecological law?*
- March 20-23, 2024. Biology Across Scales Symposium, Instituto Gulbenkian de Ciência.
  Invited Talk: *High-dimensional microbial community dynamics*
- March 5, 2024. APS March Meeting 2024.
  Invited Talk: *Microbial community dynamics in-silico, in-vitro, and in-vivo*
- March 1, 2024. Physics of Living System.
  Invited Talk: *Competition without contingency and functional convergence*
- October 19, 2023. Workshop on New Frontiers in Modelling Ecology and Evolution of Microbiomes, Friedrich Schiller University.
  Invited Talk: *Non-catastrophic shifts in the human gut microbiome*
- September 4 - September 6, 2023. Ecological networks from theory to application, Ben-Gurion University of the Negev.
  Invited Talk: *Deep Dive on linking models and data*
- June 15 - June 16, 2023. Ecological perturbations across systems and scales, University of Granada.
  Invited Talk: *Non-catastrophic shifts in the human gut microbiome*
- May 3 - May 5, 2023. Cosentino-Lagomarsino and Ciandrini group retreat,.
  Invited Talk: *Coarse-graining microbial ecology*
- October 20, 2022. EcoIncontros, Department of Ecology, USP.
  Invited talk: *What is typical in microbial communities?*
- September 14 - September 16, 2022. Workshop Mathematical modelling of microbiomes, MPI for Evolutionary Biology.
  Keynote Talk: *What is typical in microbial communities?*
- April 24, 2022. Area Science Park.
  Invited Talk: *What is typical in microbial communities?*
- April 7, 2022. Spring workshop on Physics of Data.
  Invited Talk: *What is typical in microbial communities?*
- September 18, 2020. EESB seminars, MIT, US.
  Talk: *What is typical in microbial communities?*
- September 18, 2020. Toponet 2020, NetSci 2020.
  Invited Talk: *Higher-order interactions in ecological systems*
- August 25, 2020. Theory and Modeling of Living System symposium, Emory College.
  Invited Talk: *Laws of diversity and variation in microbial communities*
- January 20-25, 2020. Community Ecology: from patterns to principles, ICTP-SAIFR.
  Invited Talk: *Laws of diversity and variation in microbial communities*
- December 9-10, 2019. Quantitative Methods in Gene Regulation V, Institute of Physics.
  Invited Talk: *Laws of diversity and variation in microbial communities*
- November 12-13, 2019. Cognitive Regime Shifts II (Working Group), Santa Fe Institute.
  Invited:
- September 2-6, 2019. Model-guided data science (School), Villa del Grumello.
  Invited Talk: *Laws of diversity and variation in microbial communities*

- August 18-23, 2019. Out-of-Equilibrium Processes in Evolution and Ecology (BIRS Workshop), Casa Matemática Oaxaca.
  Invited Talk: *Macroecological laws across microbial communities*

- July 30, 2019. SBP-Seminar, SISSA.
  Invited Talk: *Laws of diversity and variation in microbial communities*

- July 1-3, 2019. Italian Regional Conference on Complex Systems, Fondazione Bruno Kessler.
  Contributed Talk: *Macroecological laws across microbial communities*

- February 15, 2019. Pyeong Chang Forum 2019.
  Invited talk: *Mysteries and Laws of Biodiversity*

- February 11, 2019. Seoul National University.
  Invited talk: *Higher-order interactions stabilize dynamics in competitive network models*

- December 14, 2018. Department of Ecology, USP.
  Invited talk: *Higher-order interactions stabilize the dynamics of ecological communities*

- December 13, 2018. ICTP-SAIFR.
  Invited talk: *Higher-order interactions stabilize the dynamics of ecological communities*

- September 26, 2018. ReAct 3.0 @ CCS 2018.
  Invited talk: *Higher-order interactions stabilize dynamics in competitive network models*

- July 23-25, 2018. Cognitive Regime Shifts I (Working Group), Santa Fe Institute.
  Invited talk: *On the stability of large ecological communities*

- May 7-11, 2018. Statistical physics of cells and genomes (Working group).
  Invited talk: *Diversity in ecological communities*

- March 5-9, 2018. APS March Meeting 2018.
  Contributed talk: *Statistical physics of (meta)genomes*

- May 2, 2017. ICTP.
  Invited talk: *Higher-order interactions stabilize the dynamics of ecological communities*

- February 27, 2017. Second Science of Science Meeting.
  Invited talk: *What's in a Last Name? Mobility, Gender Imbalance and Nepotism across Academic Systems*

- January 26, 2017. Santa Fe Institute.
  Invited talk: *Higher-order interactions stabilize the dynamics of ecological communities*

- August 11, 2015. 100th ESA Conference.
  Contributed talk: *Feasibility and stability of large ecosystems*

- August 11, 2015. Granada Seminar.
  Contributed talk: *Persistence of a population in randomly fragmented landscapes*

- December 18, 2014. Workshop on Physics of Complex Systems.
  Invited talk: *Emergence of criticality in communities of living systems*

- September 22, 2014. ECCS 2014.
  Contributed talk: *Persistence of a population in randomly fragmented landscapes*

- September 15, 2013. ECCS 2013.
  Contributed talk: *Emergence of criticality in living systems through adaptation and evolution*

- July 1, 2013. Workshop on Quantitative Laws of Genome Evolution.
  Contributed talk: *Universal properties of ecological interactions and stability of ecosystems*

- December 20, 2012. Workshop on Physics of Complex Systems.
  Invited talk: *Growth or Reproduction? Emergence of a Strategy*
- November 9, 2012. Scientific day in honor of Bruno Bassetti.
  Invited talk: *Growth or Reproduction? Emergence of a Strategy*
- June 22, 2012. XVII Conference on Statistical Physics and Complex Systems.
  Contributed talk: *Spatial distribution of species across scales*

Scientific visits

- November 18, 2013 to May 30, 2014
  Visiting Student at Department of Ecology and Evolution, The University of Chicago, Chicago, IL, USA.
- July 22, 2013 to August 3, 2013
  Visiting Student at Departamento de Electromagnetismo y Física de la Materia, Universitad de Granada, Granada, Spain.
- February 20, 2012 to March 31, 2012
  Visiting Student at Genomic Physics Group, Genomique des Microorganismes, UMR 7238 CNRS - Université Pierre et Marie Curie, Paris, France.
- June 1, 2010 to June 28, 2010
  Summer Internship under the supervision of S. Maslov at Department of Condensed Matter Physics, Brookhaven National Laboratory, Upton, NY, USA.

Publications

[1] F. Zimmaro, J. Grilli, M. Galesic, A.J. Stewart. Cultural tightness and social cohesion under coevolving beliefs, behaviors, and preferences *Proceedings of the National Academy of Sciences*. 123(14):e2525139123. 2026.
doi:10.1073/pnas.2525139123 arXiv:2509.01848

[2] S. Golmohammadi, M. Zarei, J. Grilli. Modeling Spatial Synchronization of Predator-Prey Oscillations via the XY Model under Demographic Stochasticity and Migration *Physical Review E*. 2026.
doi:10.1103/78jb-mtnq arXiv:2511.13501

[3] R. Droghetti, P. Fuchs, I. Iuliani, V. Firmano, G. Tallarico, L. Calabrese, J. Grilli, B. Sclavi, L. Ciandrini, M. Cosentino Lagomarsino. Incoherent feedback from coupled amino acids and ribosome pools generates damped oscillations in growing E. coli *Nature Communications*. 16, 3063. 2025.
doi:10.1038/s41467-025-57789-4 pmid:40157904 bioRxiv:10.1101/2023.10.25.563923

[4] W.R. Shoemaker, J. Grilli. The macroecological dynamics of sojourn trajectories in the human gut microbiome *mSystems*. 2026.
doi:10.1128/msystems.01221-25 bioRxiv:10.1101/2025.06.27.661930

[5] W.R. Shoemaker, A. Sanchez, J. Grilli. Macroecological laws in experimental microbial systems *PLOS Computational Biology*. 21 (5), e1013044. 2025.
doi:10.1371/journal.pcbi.1013044 bioRxiv:10.1101/2023.07.24.550281

[6] S. Golmohammadi, M. Zarei, J. Grilli. The effect of demographic stochasticity on predatory-prey oscillations *The European Physical Journal Plus*. 140 (8), 781. 2025.
doi:10.1140/epjp/s13360-025-06713-2 arXiv:2310.20575

[7] W.R. Shoemaker, J. Grilli. Investigating macroecological patterns in coarse-grained microbial communities using the stochastic logistic model of growth *eLife*. 12:RP89650. 2024.
doi:10.7554/eLife.89650.3 bioRxiv:10.1101/2023.03.02.530804

[8] E. Omodei, J. Grilli, M. Marsili, G. Sanguinetti. Quantitative Human Ecology: Data, Models and Challenges for Sustainability *Quantitative Sustainability: Interdisciplinary Research for Sustainable Development Goals [book chapter]*. 2024.

[9] S. Pompei, E. Bella, J.S. Weitz, J. Grilli, M. Cosentino Lagomarsino. Metacommunity structure preserves genome diversity in the presence of gene-specific selective sweeps under moderate rates of horizontal gene transfer *Plos Computational Biology*. 19 (10), e1011532. 2023.
doi:10.1371/journal.pcbi.1011532 bioRxiv:10.1101/2023.01.17.524383

[10] M. Sireci, M.A. Muñoz, and J. Grilli. Environmental fluctuations explain the universal decay of species-abundance correlations with phylogenetic distance *Proceedings of the National Academy of Sciences*. 120 (37) e2217144120. 2023.
doi:10.1073/pnas.2217144120 pmid:37669363 bioRxiv:10.1101/2022.07.12.499693

[11] A. Mazzolini, J. Grilli. Universality of evolutionary trajectories under arbitrary forms of self-limitation and competition *Physical Review E*. 108, 034406. 2023.
doi:10.1103/PhysRevE.108.034406 bioRxiv:10.1101/2021.06.17.448795

[12] L. Fant, O. Mazzarisi, E. Panizon, J. Grilli. Stable cooperation emerges in stochastic multiplicative growth *Physical Review E*. 108, L012401. 2023.
doi:10.1103/PhysRevE.108.L012401 pmid:37583239 arXiv:2202.02787

[13] J. Grilli. Keystone intransitive loops *Proceedings of the National Academy of Sciences*. 120 (17) e2304170120 [commentary]. 2023.
doi:10.1073/pnas.2304170120 pmid:37068257

[14] I. Macocco, A. Glielmo, J. Grilli, A. Laio. Intrinsic dimension estimation for discrete metrics *Physical Review Letters*. 130, 067401. 2023.
doi:10.1103/PhysRevLett.130.067401 pmid:36827575 arXiv:2207.09688

[15] R.E. Szabo, S. Pontrelli, J. Grilli, J.A. Schwartzman, S. Pollak, U. Sauer, O.X. Cordero. Historical contingencies and phage induction diversify bacterioplankton communities at the microscale *Proceedings of the National Academy of Sciences*. 119 (30) e2117748119. 2022.
doi:10.1073/pnas.2117748119 pmid:35862452 bioRxiv:10.1101/2021.09.27.461956

[16] M. Cosentino Lagomarsino, G. Pacifico, V. Firmano, E. Bella, P. Benzoni, J. Grilli, F. Bassetti, F. Capuani, P. Cicuta, and M. Gherardi. Remote teaching data-driven physical modeling through a COVID-19 data challenge *European Journal of Physics*. 2022.
doi:10.1088/1361-6404/ac79e1 arXiv:2104.09394

[17] L. Calabrese, J. Grilli, M. Osella, C.P. Kempes, M. Cosentino Lagomarsino, L. Ciandrini. Protein degradation sets the fraction of active ribosomes at vanishing growth *Plos Computational Biology*. 18(5): e1010059. 2022.
doi:10.1371/journal.pcbi.1010059 pmid:35500024 bioRxiv:10.1101/2021.03.25.436692

[18] S. Zaoli, J. Grilli. The stochastic logistic model with correlated carrying capacities reproduces beta-diversity metrics of microbial communities *Plos Computational Biology*. 18(4):e1010043. 2022.
doi:10.1371/journal.pcbi.1010043 pmid:35363772 bioRxiv:10.1101/2021.11.16.468765

[19] F. Buke, J. Grilli, M. Cosentino Lagomarsino, G. Bokinsky, S.J. Tans. ppGpp is a bacterial cell size regulator *Current Biology*. 32(4):870-877. 2022.
doi:10.1016/j.cub.2021.12.033 pmid:34990598 bioRxiv:10.1101/2020.06.16.154187v1

[20] F. de Castro, S.M. Adl, S. Allesina, R.D. Bardgett, T. Bolger, J.J. Dalzell, M. Emmerson, T. Fleming, D. Garlaschelli, J. Grilli, S.E. Hannula, F. de Vries, Z. Lindo, A.G. Maule, M. Öpik, M.C. Rillig, S.D. Veresoglou, D.H. Wall, T. Caruso. Local stability properties of complex, species-rich soil food webs with functional block structure *Ecology and Evolution*. 00:1–12 (2021). 2021. doi:10.1002/ece3.8278 pmid:34824812

[21] S. Zaoli, J. Grilli. A macroecological description of alternative stable states reproduces intra-and inter-host variability of gut microbiome *Science Advances*. 7 (43), eabj2882. 2021. doi:10.1126/sciadv.abj2882 pmid:34669476 bioRxiv:10.1101/2021.02.12.430897

[22] L. Descheemaeker, J. Grilli, S. de Buyl. Heavy-tailed abundance distributions from stochastic Lotka-Volterra models *Physical Review E*. 104, 034404. 2021. doi:10.1103/PhysRevE.104.034404 pmid:34654137 bioRxiv:10.1101/2021.02.19.431657

[23] M. Panlilio, J. Grilli, G. Tallarico, I. Iuliani, B. Sclavi, P. Cicuta, M. Cosentino Lagomarsino. Threshold accumulation of a constitutive protein explains E. coli cell division behavior in nutrient upshifts *Proceedings of the National Academy of Sciences*. 118 (18) e2016391118. 2021. doi:10.1073/pnas.2016391118 pmid:33931503 bioRxiv:10.1101/2020.08.03.233908v1

[24] J. Grilli. Macroecological laws describe variation and diversity in microbial communities *Nature Communications*. 11, 4743. 2020. doi:10.1038/s41467-020-18529-y pmid:32958773 bioRxiv:10.1101/680454v1

[25] K. Jovic, J. Grilli, M.G. Sterken, J.A.G. Riksen, B.L. Snoek, S. Allesina, J.E. Kammenga. Transcriptome dynamics predict thermotolerance in Caenorhabditis elegans *BMC Biology*. 17, 102. 2019. doi:10.1186/s12915-019-0725-6 pmid:31822273 bioRxiv:10.1101/661652v2

[26] C. Tu, S. Suweis, J. Grilli, M. Formentin, and A. Maritan. Reconciling cooperation, biodiversity and stability in complex ecological communities *Scientific Reports*. 9, 5580. 2019. doi:10.1038/s41598-019-41614-2 pmid:30944345 arXiv:1708.03154

[27] G. Micali, J. Grilli, M. Osella, and M. Cosentino Lagomarsino. Concurrent processes set E. coli cell division *Science Advances*. 4, eaau3324. 2018. doi:10.1126/sciadv.aau3324 pmid:30417095 bioRxiv:2018/04/16/301671

[28] G. Micali, J. Grilli, J. Marchi, M. Osella, and M. Cosentino Lagomarsino. Dissecting the control mechanisms for DNA replication and cell division in E. coli *Cell Reports*. 25,3:761-771.E4. 2018. doi:10.1016/j.celrep.2018.09.061 pmid:30332654 bioRxiv:2018/04/25/308155

[29] J.N. Pruitt, A. Berdahl, C. Riehl, N. Pinter-Wollman, H.V. Moeller, E.G. Pringle, L.M. Aplin, E.J.H. Robinson, J. Grilli, P. Yeh, V.M. Savage, M.H. Price, J. Garland, I.C. Gilby, M. C. Crofoot, G.N. Doering, and E.A. Hobson. Social tipping points in animal societies *Proceedings of the Royal Society B*. 285:20181282. 2018. doi:10.1098/rspb.2018.1282 pmid:30232162

[30] T. Gibbs, J. Grilli, T. Rogers, and S. Allesina. The effect of population abundances on the stability of large random ecosystems *Physical Review E*. 98, 022410. 2018. doi:10.1103/PhysRevE.98.022410 pmid:30253626 arXiv:1708.08837

[31] C. Cadart, S. Monnier, J. Grilli, P.J. Sáez, N. Srivastava, R. Attia, E. Terriac, B. Baum, M. Cosentino Lagomarsino, and M. Piel. Size control in mammalian cells involves modulation of both growth rate and cell cycle duration *Nature Communications*. 9:3275. 2018. doi:10.1038/s41467-018-05393-0 pmid:30115907 bioRxiv:2017/06/20/152728

[32] J. Grilli, C. Cadart, G. Micali, M. Osella, and M. Cosentino Lagomarsino. The empirical fluctuation pattern of E. coli division control *Frontiers in Microbiology*. 9, 1541. 2018.
doi:10.3389/fmicb.2018.01541 pmid:30105006

[33] A. Mazzolini, J. Grilli, E. De Lazzari, M. Osella, M. Cosentino Lagomarsino, and M. Gherardi. Zipf and Heaps laws from dependency structures in component systems *Physical Review E*. 98, 012315. 2018.
doi:10.1103/PhysRevE.98.012315 pmid:30110773 arXiv:1801.06438

[34] C.A. Serván, J.A. Capitán, J. Grilli, K.E. Morrison, and S. Allesina. Coexistence of many species in random ecosystems *Nature Ecology & Evolution*. 2, 1237–1242. 2018.
doi:10.1038/s41559-018-0603-6 pmid:29988167

[35] K. Jovic, M.G. Sterken, J. Grilli, R.P.J. Bevers, M. Rodriguez, J.A.G. Riksen, S. Allesina, J.E. Kammenga, and B.L. Snoek. Temporal dynamics of gene expression in heat-stressed Caenorhabditis elegans *Plos One*. 2(12), e0189445. 2017.
doi:10.1371/journal.pone.0189445 pmid:29228038 bioRxiv:2017/05/16/135988

[36] J. Grilli, G. Barabás, M.J. Michalska-Smith, and S. Allesina. Higher-order interactions stabilize dynamics in competitive network models *Nature*. 548, 210–213. 2017.
doi:10.1038/nature23273 pmid:28746307

[37] J. Grilliand S. Allesina. Last name analysis of mobility, gender imbalance, and nepotism across academic systems *Proceedings of the National Academy of Sciences*. 114(29):7600-7605. 2017.
doi:10.1073/pnas.1703513114 pmid:28673985

[38] C. Tu, J. Grilli, F. Schuessler, and S. Suweis. Collapse of resilience patterns in generalized Lotka-Volterra dynamics and beyond *Physical Review E*. 95, 062307. 2017.
doi:10.1103/PhysRevE.95.062307 pmid:28709280 arXiv:1606.0963

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arXiv:1407.2425

Teaching Experience

April-May 2020, 2021, 2022, 2023
Scientific Storytelling and Critical Thinking. Diploma in Quantitative Life Science, ICTP (16 hours).

November-December 2020, 2021
Communication of epidemics. (with R. Villa) Master in Comunicazione della Scienza, SISSA (10 hours).

October-December 2019, 2020, 2021, 2022
Introduction to Ecology and Evolution. Diploma in Quantitative Life Science, ICTP and Master in Physics of Complex Systems (52 hours).

November 2017
Advanced topics in stochastic processes - Random Matrix Theory (with A. Maritan & S. Suweis). Ph.D. School in Physics, Università degli Studi di Padova. (8 hours)

9 November 2017
Lecture on *neutral theory* during the class *An Introduction to Stochastic Processes in Continuous Time* (held by D. Alonso). Ph.D. program in Ecology&Evolution, University of Chicago. (2 hours)

October 2014
Introduction to Complex Systems (with S. Suweis). Master in Scientific Communication, Università degli Studi di Padova. (2 hours)

September 2014
Tutor at ESTAGE, internship for high-school students at Department of Physics and Astronomy, Università degli Studi di Padova. (8 hours)

November 2012 - June 2013
Tutor Junior at Università degli Studi di Padova
Mathematics (for 1st year Geology students), Mathematical Analysis and Linear Algebra (for 1st year Physics students).

Supervision Current

- W.R. Shoemaker, Postdoc 2022-, ICTP, Trieste, Italy.
- R. Crisostomo, Master Student, 2023-,QBio at U. of Montpellier, France
- X. Zhoum, Visiting Ph.D. Student, 2022-, HHU, China
- S. Golmohammadi, Ph.D. Student, 2019-present, IASBS, Iran and STEP Program, ICTP, Italy (with M. Zarei)

Supervision Past

- J. M. Camacho Mateu, Visiting Ph.D. Student 2023, Carlos III University of Madrid, Madrid, Spain.
- V. Monteiro Marquioni, Visiting Ph.D. Student 2022-2023, Universidade Estadual de Campinas, Campinas, Brazil.
- S. Lipani, M.Sc. Student 2022, University of Padova, Padova, Italy.
- M. Sireci, Ph.D. Student 2019-2023, University of Granada, Granada, Spain (with M.A. Munõz).
- M. Vasquez Ibarra, QLS Diploma Student 2021-2022, ICTP, Trieste, Italy.
- S. Farrag, QLS Diploma Student 2021-2022, ICTP, Trieste, Italy.
- S. Zaoli, Postdoc 2020-2022, ICTP, Trieste, Italy.
- L. Fant, Ph.D. Student 2019-2022, SISSA and ICTP, Trieste, Italy.
- A. Valsecchi, M.Sc. Student 2021-2022, University of Milan, Milan, Italy.
- A. Saleem, QLS Diploma Student 2020-2021, ICTP, Trieste, Italy.
- I. Rondon, QLS Diploma Student 2020-2021, ICTP, Trieste, Italy.
- M.H. Ming, MATH Diploma Student 2020-2021, ICTP, Trieste, Italy.
- M. Corigliano, M.Sc. Student 2020-2021, University of Milan, Milan, Italy.
- N. Dorilas, Research Experience for Undergraduates 2018, Santa Fe Institute, US (with A. Rominger).

- T. Gibbs, Undergraduate Student 2016-2017, Ecology & Evolution, Chicago, US (with S. Allesina).
- R. Satterwhite, Ph.D. Student (rotation) 2015, Ecology & Evolution, Chicago, US (with S. Allesina).
- M. Adorisio, M.Sc. in Physics 2014, Padova, Italy (with A. Maritan and S. Suweis).
- M. Insolia, B.Sc. in Physics 2014, Padova, Italy (with A. Maritan).
- E. De Lazzari, M.Sc. in Physics 2013, Padova, Italy (with A. Maritan and S. Suweis).

Ph.D. panels and commitees

- Malyon Bimler, Ph.D. in Ecology, University of Queensland, Australia (2020)
- Lana Descheemaeker, Ph.D. in Applied Physics, Vrije University of Brussels, Belgium (2020)
- Samuele Stivanello, Ph.D. in Mathematics, University of Padova, Italy (2021)
- Leonardo Miele, Ph.D. in Mathematics, University of Leeds, UK (2021)
- Rafael Menezes, Ph.D. program in Ecology, Universidade de São Paulo, Brazil (2021-)
- Francesco Camaglia, Ph.D. in Physics, LPENS, Paris, France (2023)

Grants, Fellowships and Awards

- July 2019
  css/italy young scientist award
- January 2018 to December 2019
  Omidyar Fellowship, Santa Fe Institute.
- January 2014
  Fellowship sponsored by the Ing. Aldo Gini private foundation in Padua, funding a visit of 6 months at the University of Chicago [4.8k€].
- January 2012 to December 2014
  Three years fellowship for Ph.D. studies from Università degli Studi di Padova.
- October 2011 to December 2011
  Post-master scholarship 'ex 60%' 2011.

Habilitations

- August 8, 2018 to August 8, 2028
  Italian National Scientific Habilitation as Associate Professor in Theoretical Physics of Matter (ASN, 02/B2 II Fascia).
- September 12, 2018 to September 12, 2028
  Italian National Scientific Habilitation as Associate Professor in Applied Physics (ASN, 02/D1 II Fascia).
- October 8, 2018 to October 8, 2028
  Italian National Scientific Habilitation as Associate Professor in Ecology (ASN, 05/C1 II Fascia).

Other

Languages
Italian (native speaker), English (fluent) and Spanish (good)

Member of American Physical Society (2014,2018) Member of Ecological Society of America (2015), Member of Complex System Society (2013-2014),

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